Chinese Characters as Reactions: Typesetting Decompositions with `zitex`

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Abstract

We write the decomposition of a Chinese character in a few lines of data, and `zitex` turns it into a tree, a nested Zi-Matrix, a one-line formula and a typeset equation. This file is both a short paper and a guide: each block of markup is followed by a note on its syntax, and the features it shows.

1 Introduction

A character can be read as the product of a reaction, for example 亻 + 弗 → 佛. Writing that by hand in every paper is slow, and the layout of the parts (left–right, top–bottom, enclosing) is lost. With `zitex` the same thing in the middle of a sentence is 亻 + 弗 → **佛** (fó, Buddha), and the layout is kept in the matrix of Figure 1.

Syntax. `\ziformula{字}` prints *parts* → *character (pinyin, gloss)* on one line, so it works inside a sentence, a caption or a table cell. It is the only style that is not a block.

The data. A character is a short YAML entry: its shape in prefix form (for 佛, 亻 at `L` and 弗 at `M`), plus the position of each part in its level. The positions are the eleven Zi-Matrix codes `L R M U D Mi Mo LU LD RU RD`, where `M` is the centre of gravity and a Kangxi radical is peripheral. This paper uses three characters, taken from a database with `zitex extract`:

Character	Pinyin	Meaning	First-level pattern	First-level parts
器	qì	device	`LU-RU-M-LD-RD`	口口犬口口
藻	zǎo	aquatic grasses	`U-D`	艹 + 澡
佛	fó	Buddha	`L-M`	亻 + 弗

Syntax. The long form `\zi[style=formula]{器}` and the short form `\ziformula{器}` are the same. Every style has a shortcut: `\zitree`, `\zimatrix`, `\ziformula`, `\zimath`. The parts are read from `chars.yaml`, and the table is typed by hand.

2 Four views of one decomposition

2.1 Equation: `\zimath`

The equation view is the most compact. Pinyin sits above each part, and the gloss below.

$$\underbrace{\overset{\text{cǎo}}{艹}}_{\text{grass}} + \underbrace{\overset{\text{zǎo}}{澡}}_{\text{bath}} = \underbrace{\overset{\text{zǎo}}{藻}}_{\text{aquatic grasses}}$$

*Every figure, equation and formula below comes from `\zi`-commands. The text around them is ordinary LaTeX.

Syntax. `\zimath{字}` is a display equation made only of `\text`, `\overset`, `\underbrace` and `\xrightarrow`, so the same snippet works in MathJax. A part that has parts of its own is written as a bracketed group, which is how 藻 shows 澡 inside it (see the tree and the matrix below). The `lang` key picks the gloss language, as in `\zimath[lang=fr]{悔}`, and needs that language in the lexicon.

2.2 Matrix: `\zimatrix`

The matrix keeps the *geometry*: parts are boxes placed where the position codes say.

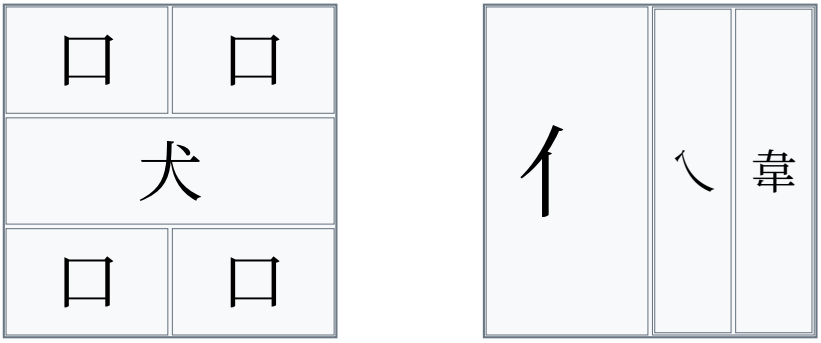


Figure 1: Zi-Matrices of 器 (left: four 口 at the corners and 犬 in the centre) and 佛 (right).

Syntax. `\zimatrix{字}` draws with TikZ and is an ordinary box, so it goes in a `figure`, a table cell, or a `\resizebox`. Two matrices fit side by side here because each is narrow. The default text size is set by the snippet. To change it, scale the box: `\resizebox{0.8\linewidth}{!}{\zimatrix{器}}`.

2.3 Tree: `\zitree`

The tree shows how deep a decomposition goes. 藻 has four levels, because 澡 and 喿 are themselves decomposed.

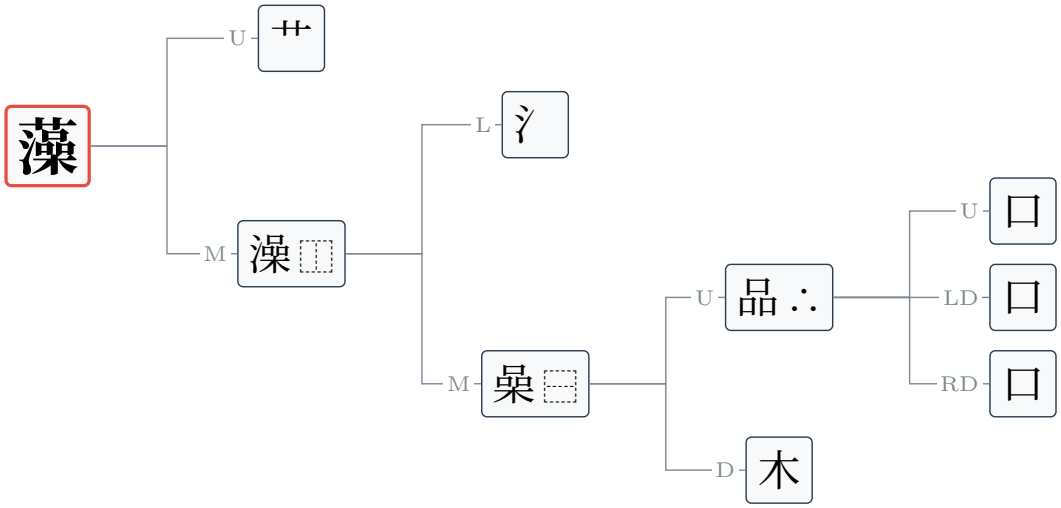


Figure 2: The decomposition tree of 藻. Leaves are the parts that are not decomposed further.

Syntax. A tree can be wide, so Figure 2 is wrapped in `\resizebox`. The wrapper only scales, the snippet itself is unchanged. `\zitree` accepts the same keys as the others: `style`, `lang`, `reading`.

2.4 Choosing the style as an option

All four commands are shortcuts for one: `\zi[style=...]`.

Command	Output for 佛	Use it for
`\zimath{佛}`	an equation (display)	the main text, one idea per equation
`\zimatrix{佛}`	nested boxes	figures, showing the layout
`\zitree{佛}`	a tree	figures, showing the depth
`\ziformula{佛}`	亻 + 弗 → **佛** (fó, Buddha)	a sentence, a caption, a table

3 Results: a small gallery

The same three characters as a matrix and as a formula:

Syntax. Block styles work in a `tabular` as long as the column is wide enough, which is why the matrices are scaled with `\resizebox`. `\ref`, `\label` and `\caption` behave as for any figure or table, so a paper can refer to Table 1, Figure 1 and Figure 2 in the usual way.

Observation. In 器, a dog (犬) sits inside four mouths (口). In 藻 the grass radical (艹) is on top and the water radical (氵) is a part of 澡, which is why the equation reads 艹 + 澡 → **藻** (zǎo, aquatic grasses) and the tree goes deeper (Figure 2). This is a description of the data, not a claim about etymology.

4 Features you will want in a real paper

4.1 Language and reading

Equations and formulas print glosses, so they depend on a language. Set a default for the document or override it in one call:

```
\zitexsetlang{fr}                % glosses in French from here on
\zimath[lang=en]{藻}             % one call in English
\zitexsetreading{dictionary}     % use a named reading
\zimath[reading=author]{虹}      % one call with another reading
```

Syntax. A *reading* is a named interpretation of the roles of the parts (semantic, phonetic, force), so two schools of thought can sit in the same data and a paper can choose. In the file system they appear as `<char>-<style>[-<lang>][@reading].tex`. These lines are not run here, because the lexicon of this folder has English only and no readings.

4.2 Where the snippets are

```
\zitexsetpath{zitex-out}         % the default: a folder next to main.tex
```

Syntax. To share one folder between several papers, point every paper to it, and run `zitex extract -o` on that folder.

4.3 Fonts

The `zitex` package has a font fallback: when the primary font has no glyph for a part (rare characters, ancient components), a fallback font, an SVG image or an alias is used, as set in `fonts.yaml`. A part that nothing can draw prints a boxed code point such as `U+XXXX` and is listed by `zitex fonts`.

5 Reproducing this paper

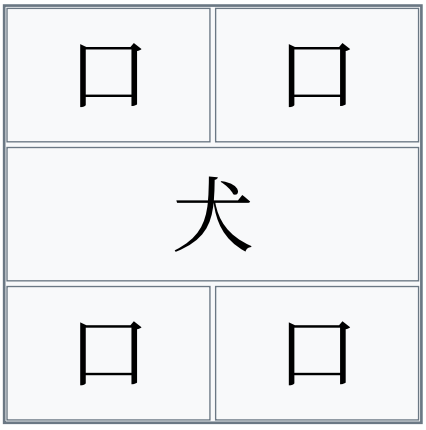
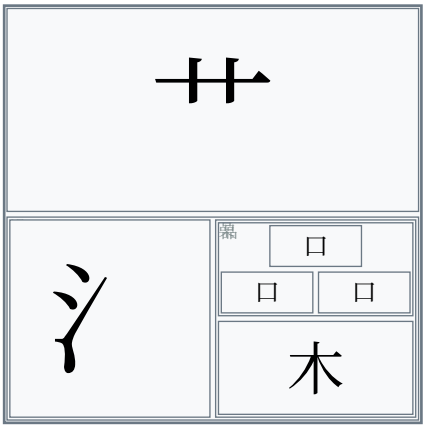
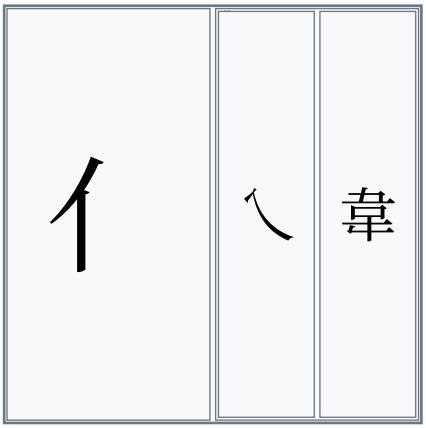
```
zitex setup   -o work/                          # zitex.sty, fonts.yaml, glyphs/
zitex extract -c 器,藻,佛 -o work/             # chars.yaml, lexicon.yaml, main.tex, zitex-out/
cd work && xelatex main.tex                     # the PDF
```

Syntax. `extract` writes a minimal `main.tex` and never replaces one that exists unless you pass `--force`. To use this file instead, copy it over `main.tex`, and add any new character to the `-c` list. If a character is missing, the red mark and the log line `Missing zitex-out/...` say which.

Summary of the syntax.

Markup	Meaning
`\usepackage{zitex}`	load the package (needs XeLaTeX)
`\zi[style=S, lang=L, reading=R]{字}`	general form; `S` is one of tree, matrix, formula, math
`\zitree`, `\zimatrix`, `\ziformula`, `\zimath`	shortcuts for the four styles
`\zitexsetlang{L}`, `\zitexsetreading{R}`	document-wide defaults
`\zitexsetpath{dir}`	where the snippets are (default `zitex-out`)

Table 1: One row per character. Cells hold `\zi` output directly.

	Matrix	Formula
器		口 + 口 + 犬 + 口 + 口 → **器** (qì, device)
藻		艹 + 澡 → **藻** (zǎo, aquatic grasses)
佛		亻 + 弗 → **佛** (fó, Buddha)